Quantum Secure Threshold Private Set Intersection Protocol for IoT-Enabled Privacy-Preserving Ride-Sharing Application

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***Abstract*—The Internet of Things (IoT)-enabled ride sharing is one of the most transforming and innovative technologies in the transportation industry. It has myriads of advantages, but with increasing demands there are security concerns as well. Traditionally, cryptographic methods are used to address the security and privacy concerns in a ride sharing system. Unfortunately, due to the emergence of quantum algorithms, these cryptographic protocols may not remain secure. Hence, there is a necessity for privacy-preserving ride sharing protocols which can resist various attacks against quantum computers. In the domain of privacy-preserving ride sharing, a threshold private set intersection (TPSI) can be adopted as a viable solution because it enables the users to determine the intersection of private data sets if the set intersection cardinality is greater than or equal to a threshold value. Although TPSI can help to alleviate privacy concerns, none of the existing TPSI is quantum secure. Furthermore, the existing TPSI faces the issue of long-term security. In contrast to classical and post quantum cryptography, quantum cryptography (QC) provides a more robust solution, where QC is based on the postulates of quantum physics (e.g., Heisenberg uncertainty principle, no cloning theorem, etc.) and it can handle the prevailing issues of quantum threat and long-term security. Herein, we propose the *first* QC-based TPSI protocol which has a direct application in privacy-preserving ride sharing. Due to the use of QC, our IoT-enabled ride sharing scheme remains quantum secure and achieves long-term security as well.**

***Index Terms*—Internet of Things (IoT), long-term security, private set intersection (PSI), quantum communication, ride-sharing.**

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I. Introduction

THE Internet of Things (IoT) is a cutting-edge technology that will connect numerous physical things that communicate without the need for human contact. In technical terms, IoT is a network of connected computing devices, mechanical and digital machinery, items, animals, or people that may exchange data across a network without needing human-to-human or human-to-computer communication. Thus, it is expected that IoT will be a crucial part of our daily lives in the coming years. IoT offers a wide range of applications in every aspect of our lives.

The IoT's conveniences come with new security dangers and privacy concerns that need to be appropriately handled. The key challenges in an IoT environment are concerns, such as privacy, authorization, verification, access control, system setup, information storage, and administration. IoT being so pervasive in day-to-day human life makes it a great risk to users' privacy. Ignoring these privacy and security concerns would negatively impact many elements of our life, including our houses and the vehicles we use to get to work, and our bodies. IoT privacy protection is essential because an outsider may learn a lot about a person's life by listening in on the sensed data that their wearable technology and smart home gadgets relay. Even if IoT devices just report metadata, it is possible to gather a significant quantity of data about a user's daily life by combining the metadata from several compromised things that surround him/her over time. As a result, security and privacy concerns must be resolved in order to provide users confidence in their privacy and control over their personal information before the IoT is widely used.

In the transportation industry, IoT-enabled ride sharing plays an important role due to its wide advantages like reduced travel expenditure, reduced congestion, etc. It further helps in reducing the pollution. Despite its advantages, several roadblocks have restricted its widespread adoption. For instance, the current technique used in ride sharing is centralized [1] and the service provider companies have access to passengers data. Thus, if ride sharing is put in practice without proper assessment, it may turn out to be a tool for damaging and compromising the passenger's privacy. As a consequence, privacy-preserving ride sharing [2] becomes essential where two users only want to share a ride if large parts of their trajectories on a map intersect. In this case, the users may be

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interested in the concrete intersection of their routes, but only if the overlap is large. Over the years, several proposals have been given for privacy-preserving ride sharing in [3], [4], [5], [6], and [7]. Among these, threshold private set intersection (TPSI) seems to be the most suitable candidate to address the problem. TPSI is a two party cryptographic protocol that allows the entities to obtain the intersection of their private sets only when the cardinality of the intersection is greater than or equal to a threshold value. In TPSI, nothing beyond the intersection is revealed to the recipients. In the current state-of-the-art, many designs of TPSI [4], [8], [9], [10], [11], [12], [13], [14] have been proposed. Security of nearly all of them rely on the number theoretical hardness assumptions. With the advent of quantum algorithms, like Shor's algorithm [15], these hard problems are no longer secure in the quantum domain. Also, the existing TPSI protocols do not address the problem of long-term security. This makes the requirement of quantum computer immune TPSI protocols which can attain long-term security. Quantum cryptography (QC) seems to be an ideal choice to address these issues, because unlike classical cryptography (which is based on hardness of number theoretic problems), QC is immune from quantum computing attacks (e.g., Shor's algorithm [15]). In addition, it provides long-term security. Therefore, QC is an alternative to provide security and privacy in an IoT setting. The security of QC depends upon the principles of quantum mechanics. QC is primarily based on the Heisenberg's uncertainty principle and the principle of photon polarization. The security of QC-based cryptographic primitives is based on the inviolability of the aforementioned principles. In our proposed design, we make use of the part of QC which uses these two principles to build secure cryptographic primitives.

A. Motivation

The following are the main motivations behind the proposal of the scheme in this article.

1) Private set intersection (PSI) is one of most important primitives for secure computation. As already discussed, current protocols for PSI are based on number-theoretic hardness assumptions, and hence they would not be secure in the future due to attacks through quantum algorithms [15]. Motivated by this, researchers have designed several protocols for PSI in the quantum domain [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27]. However, in certain scenarios, the standard PSI functionality is not enough. In particular, it may happen that the clients may only be agreeing to reveal the intersection only if there is a large intersection. This functionality may be achieved through TPSI. *However, there is no construction of secure TPSI protocol in the quantum domain.*
2) The security of nearly all of the existing state-of-the-art TPSI relies on the number-theoretical hardness assumptions. The emergence of quantum computing makes these protocols obsolete and insecure. There is a lack of TPSI protocols which can resist these attacks by quantum computers. Hence, it is high time to design a quantum resistant protocol to achieve the TPSI functionality.
3) IoT-enabled ride sharing is one of the fastest growing industries. Use of IoT in the ride-sharing sector can bring disruptive changes. External data regarding the traffic situation, traffic lights, etc., can be used to address issues like congestion and jams in the cities. Data collected through IoT sensors can also help in providing a better user experience as these information can be factored while assigning cars, estimating time, and expenditure of travel etc. *However, there is no IoT-enabled ride sharing scheme that can resist quantum attacks.*
4) With Shor's algorithm [15], number-theoretic cryptographic schemes can be broken. Almost all of the currently used schemes [28], [29], [30], [31], [32], [33], [34] used to provide security and privacy in an IoT setting [35] are based on the number theoretic problems. Hence, security of all such IoT systems is under a potential threat [36]. As of the year 2022, some organizations have built quantum computers with up to 50 qubits [37]. Big companies are investing a lot on quantum hardware as well [38]. In addition, there has been development of software development kits (SDKs) which can be used by anyone to work with quantum computers. There are also cloud services that can be used for running a code on a real quantum computer. Note that even one quantum computer is enough to mount a full scale quantum attack on an IoT system. To address the threats posed by quantum computing attacks (like Shor's algorithm), cryptographic research community has started working in a new direction of research known as "post QC (PQC)." The security of PQC depends on mathematical problems which are hard to solve even by a quantum computer. Over the last five years, a lot of work has been done in the context of PQC-based cryptographic primitives for IoT systems [36], [39], [40], [41]. Although PQC is an exciting alternative, it suffers from the fact that it does not provide long-term security. As a result, QC provides an alternative direction of research to provide strong security guarantees, i.e., QC-based primitives provide a long term security as well as security against quantum attacks. These motivate us to design a secure and robust QC-based protocol for IoT systems.

B. Research Contributions

In this article, we focus on the design and analysis of TPSI in the context of QC. We call our proposed design as QuTPSI. Design of the proposed scheme is motivated by the works of [17] and [42]. The quantum key distribution (QKD) in [43], and I, S, and T quantum gates are used as the fundamental building blocks of our design. Security of QuTPSI is based on the basic principles of quantum mechanics. As a consequence, it is secure against quantum attacks and provides long-term security, unlike the existing works [8], [11], [44], [45]. Our design achieves the stipulated security requirements of a TPSI, namely, the trusted party cannot obtain any information about the private sets of the users, no user gets any information about the private set of other apart from the intersection, and also

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TABLE I
TECHNIQUES, ADVANTAGES, AND LIMITATIONS OF EXISTING TPSI IN TWO PARTY AND MULTIPARTY SETTING

Scheme	Cryptographic Techniques	Advantages	Drawbacks/Limitations
Kissner *et al.* [11]	* Hash Functions * Mix-Net and Shuffling Protocol * Equivocal Commitment * Additively Homomorphic Encryption	* TPSI for $n \geq 2$ honest-but-curious parties * Secure against $n-1$ dishonest colluding parties * Over threshold PSI in the malicious setting	* Does not offer long term security * Insecure against quantum attacks
Zhao *et al.* [14]	* Additive Homomorphic Encryption * Oblivious Polynomial Evaluations * Bloom Filters	* Application oriented * Server client model	* Vulnerable against quantum attacks * Does not offer long term security * Does not consider malicious adversary model
Ghosh *et al.* [8]	* Oblivious Linear Function Evaluation * Additive Homomorphic Encryption	* Started the study of sublinear TSPI * Pointed out a not yet known connections between TPSI, set reconciliation protocol, sparse polynomial interpolation	* Does not offer long term security * Insecure against quantum attacks
Badrinarayan *et al.* [10]	* Threshold Fully Homomorphic Encryption * Threshold additive homomorphic encryption with distributed setup	* Sublinear (in the set sizes) communication lower and upper bounds for TPSI * Achieve semi-honest security where up to $(n-1)$ parties could be corrupted * Protocols in star network topology	* Does not offer long term security * Insecure against quantum attacks
Mahdavi *et al.* [9].	* Shamir's Secret Sharing * Oblivious Pseudo-Random Functions * Paillier Cryptosystem	* Constant round protocol * A practical implementation and evaluation of TPSI protocol	* Does not offer long term security * Insecure against quantum attacks
Bay *et al.* [12]	* Bloom Filters and Inverted Bloom Filters * Additivelly Homomorphic Threshold PKE	* First TPSI protocol to be implemented and open-sourced * No trusted dealer required	* Does not offer long term security * Insecure against quantum attacks

an outsider is unable to obtain information about the private sets of users. Both computation and communication cost of QuTPSI is $\mathcal{O}(q + (r + r^*)q)$ where q, r, and r^* denote the size of the private vectors, number of initial photons, and number of auxiliary photons, respectively. Our proposed QuTPSI is practical and reliable with present quantum hardware technologies, since simple single-particle projective measurements and single photons are used in the design. We also investigate the probable application of our presented design QuTPSI in the context of IoT-enabled ride sharing.

C. Paper Outline

The remainder of this article is structured as follows. The related works in the domain of TPSI are discussed at length in Section II. We discuss the mathematical preliminaries in Section III. We describe the construction of our proposed quantum secure TPSI in Section IV. The correctness and security analysis are given in Sections V and VI, respectively. We compare our design with other existing TPSI in Section VII. Next, we explain how our proposed design can be used as a building block in an IoT-enabled ride-sharing stetting in Section VIII. Finally, this article is concluded in Section IX.

II. Related Work: Classical TPSI Protocols

The work of Kissner and Song [11] dealt with the problem of designing an effective protocol for TPSI and over-threshold PSI. The schemes presented in [11] employed cryptographic methods like homomorphic encryption, equivocal commitment, and mix-net and shuffling protocol, and were proven to be secure in honest-but-curious and malicious setting.

Zhao and Chow [14] designed an application oriented TPSI in the server–client model. From the efficiency point of view, the constructions presented by [14] provided linear complexity in the size of the set. Their work greatly improved the classical approach based on Garbled circuit. By utilizing the additively homomorphic encryption, Ghosh, and Simkin [8] presented the first construction and study of TPSI with sublinear complexity (in the size of the set). This was one of the first works that theoretically characterized this class of TPSI. In addition, they provided a lower bound on the complexity of sublinear TPSI. This was the first work which initiated the study of sublinear TPSI and provided a first characterization of its communication complexity.

Ghosh and Simkin [8] pointed out a not yet known connections between TPSI, set reconciliation protocol, sparse polynomial interpolation. Badrinarayanan et al. [10] extended the work of [8] and presented two functionalities for multiparty TPSI using the cryptographic primitives like homomorphic encryption. Badrinarayanan et al. [10] solved one of the open problems mentioned in the work provided in [8]. Two over threshold multiparty PSI were constructed by Mahdavi et al. [9] by utilizing cryptographic building blocks like Shamir secret sharing scheme and oblivious pseudo-random functions. They designed a constant round protocol with communication complexity of $O(nmtk)$.

Recently, Bay et al. [12] proposed two PSI protocols using bloom filter and threshold homomorphic public-key encryption schemes. Their design scaled quadratically in the number of parties involved with no requirement of a trusted dealer. The protocols presented in [12] are secure in the semi-honest model. Table I provides a summary of various techniques,

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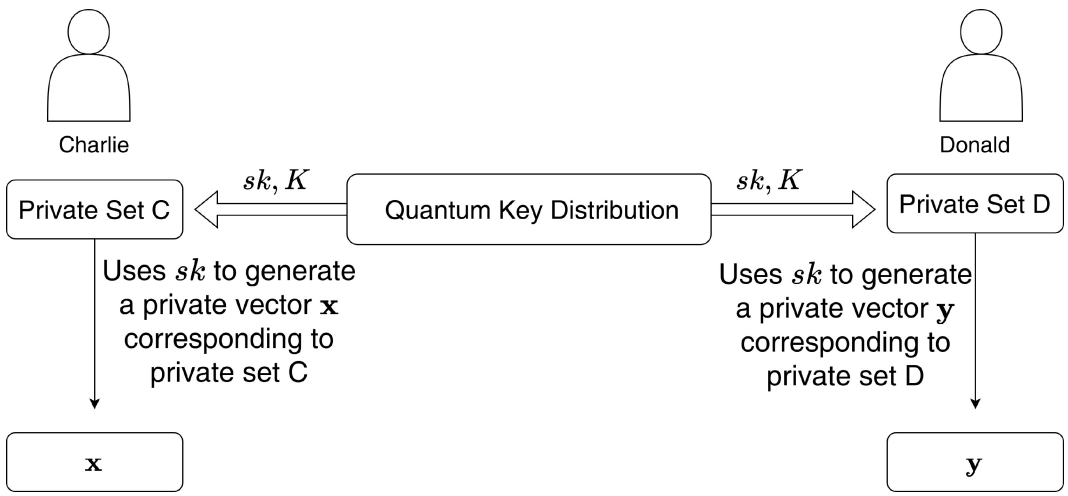

Fig. 1. Communication flow of our proposed design (Steps 1 to 3).

TABLE II
INPUT/OUTPUT TRUTH TABLES FOR S GATE

Input	Output
$\|0\rangle$	$\|0\rangle$
$\|1\rangle$	$e^{\frac{i\pi}{2}}\|1\rangle$

TABLE III
INPUT/OUTPUT TRUTH TABLES FOR T GATE

Input	Output
$\|0\rangle$	$\|0\rangle$
$\|1\rangle$	$e^{\frac{i\pi}{4}}\|1\rangle$

advantages, and limitations of existing TPSI in two party and multiparty setting.

III. MATHEMATICAL PRELIMINARIES

In this section, we discuss about the I, S, and T gates [46]. Quantum gate is a unitary operator described by a unitary matrix. More specifically, a quantum gate operating on n qubits can be represented by a unitary matrix of size $2^n \times 2^n$. For a single qubit, we require a matrix of order $2^1 = 2$. In other words, a quantum gate acting on a single qubit (also known as single qubit gate) will be a 2×2 unitary matrix. I, S, and T are three single qubit gates represented as

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, S = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix}, \text{and } T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

respectively.

Input/output truth tables for S and T gates are given in Tables II and III, respectively. Note that S and T gates perform a rotation by an angle of $\pi/2$ radian and $\pi/4$ radian, respectively, around z-axis.

From the above definition of S and T gates, we can observe that ST and TS both correspond to a rotation by an angle of $3\pi/4$ radian around z-axis with matrix representation

$$TS = ST = \begin{pmatrix} 1 & 0 \\ 0 & e^{i3\pi/4} \end{pmatrix}.$$

TABLE IV
INPUT/OUTPUT TRUTH TABLES FOR ST GATE

Input	Output
$\|0\rangle$	$\|0\rangle$
$\|1\rangle$	$e^{\frac{i3\pi}{4}}\|1\rangle$

Input/output truth tables for $ST = TS$ gate are given in Table IV.

IV. PROPOSED QUANTUM THRESHOLD PRIVATE SET INTERSECTION (QuTPSI)

In this section, we first provide a high-level overview of the proposed QuTPSI. Next, we provide the detailed description of QuTPSI.

A. High-Level Overview of QuTPSI

The QuTPSI is run between Charlie with private set $C = \{c_1, \ldots, c_l\} \subseteq \mathbb{Z}_q$ and Donald with private set $D = \{d_1, \ldots, d_m\} \subseteq \mathbb{Z}_q$. A trusted third party (TP) is also involved in this protocol for determining the set intersection cardinality by interacting with Charlie and Donald. If the cardinality is greater than or equal to a prefixed threshold value $t(\in \mathbb{N})$, then, the TP sends necessary information to Charlie and Donald for calculating the intersection. The protocol for QuTPSI is summarized in Figs. 1–3.

B. Detailed Description of QuTPSI

In the following, Charlie and Donald privately evaluate the desired intersection. We now explain our proposed QuTPSI in detail. The steps involved in QuTPSI are as follows.

1) Charlie and Donald execute a QKD [43] protocol in order to share a nonzero binary secret key sk of length greater than or equal to $\lceil \log_2(q) \rceil$ bits. Let sk correspond to $k \in \mathbb{Z}_q$. In the following, Charlie calculates $C^* = \{kc_1 \mod q, \ldots, kc_l \mod q\}$, and Donald calculates $D^* = \{kd_1 \mod q, \ldots, kd_m \mod q\}$ by using the secret integer k.

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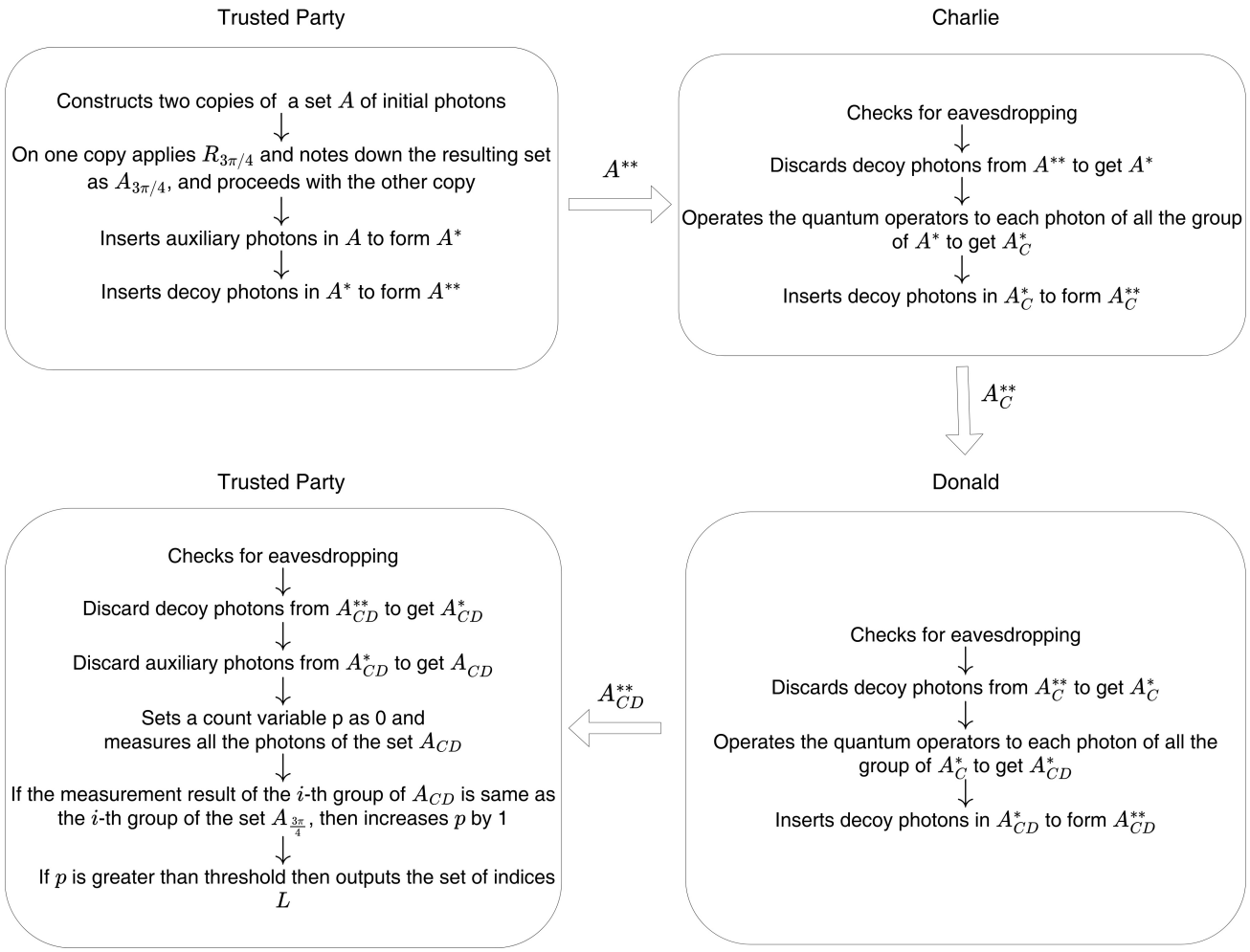

Fig. 2. Communication flow of our proposed design (Steps 4 to 9).

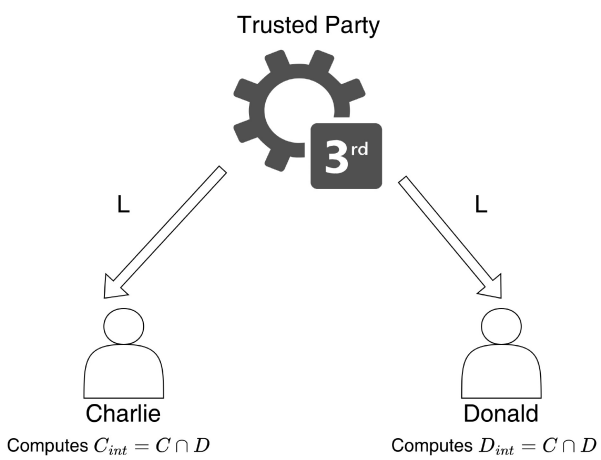

Fig. 3. Communication flow of our proposed design (Step 10).

2) Utilizing the sets C^* and D^*, Charlie and Donald generate respective private vectors $(x_0, x_1, \ldots, x_{q-1})$ and $(y_0, y_1, \ldots, y_{q-1})$ over $\mathbb{Z}_q$ in the following manner:

$$x_i = 0 \text{ if } i \notin C^*,\ x_i = 1 \text{ if } i \in C^*$$
$$y_i = 0 \text{ if } i \notin D^*,\ y_i = 1 \text{ if } i \in D^*.$$

3) Charlie and Donald share a q-bit private-key K by employing a QKD protocol [43]. The ith bit of K is considered as $K(i)$.
4) TP randomly chooses q groups of photons, where each group contains r photons (also called initial photons) having the same state from $\{|0'\rangle, |1'\rangle, |+'\rangle, |-'\rangle\}$ with $|0'\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$, $|1'\rangle = \sin\theta|0\rangle - \cos\theta|1\rangle$, $|+'\rangle = (1/\sqrt{2})(|0'\rangle + |1'\rangle)$, and $|-'\rangle = (1/\sqrt{2})(|0'\rangle - |1'\rangle)$ for $\theta \in (0, \pi/10)$. Let the set of these q groups of photons be $A = \{a_1^1, \ldots, a_r^1; a_1^2, \ldots, a_r^2; \cdots; a_1^q, \ldots, a_r^q\}$. TP makes two such copies of A, applies $R_{3\pi/4} = \begin{pmatrix} 1 & 0 \\ 0 & e^{(i3\pi/4)} \end{pmatrix}$ on one such copy and writes down the resulting set as $A_{(3\pi/4)}$. Note that TP keeps $A_{(3\pi/4)}$ with it for future and proceeds with A in the next step.
5) In each group of A, TP randomly inserts $r^* \leq r$ randomly chosen single photons (also called auxiliary photons) from $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$. The resulting set is considered as A^* which may have the form $\{a_1^1, a_1^{*1}, \ldots, a_{r^*}^{*1}, \ldots, a_r^1;\ a_1^2, \ldots, a_1^{*2}, \ldots, a_{r^*}^{*2}, \ldots a_r^2; \cdots; a_1^q, \ldots, a_{r^*}^{*2}, \ldots, a_{r^*}^{*2}, \ldots, a_r^q\}$. In addition, TP randomly inserts randomly selected l_1 decoy photons from $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ in A^* to get A^{**}. TP also notes the

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position and initial state of each photon and sends A^{**} to Charlie through a quantum channel.

6) On receiving A^{**}, Charlie first queries for l_1 decoy photon positions and corresponding measurement bases, and then announces the measurement results of each of those decoy photons in the correct bases. In the following, TP checks the eavesdropping by comparing these measurement results with the initial states of these l_1 decoy photons. If the error rate is less than or equal to the threshold value which is predetermined by them, they proceed to the next step; otherwise, they abort the process.
7) Charlie gets A^* after discarding the l_1 decoy photons from A^{**}. Now, Charlie operates the quantum operators I or S or T to each photon of all the groups of A^* in the following manner: operates I to each of the photons of the ith group if $x_{i-1} = 0$, operates S to each of the photons of the ith group if $x_{i-1} = 1$ and $K(i) = 0$, and operates T to each of the photons of the ith group if $x_{i-1} = 1$ and $K(i) = 1$. Let the resulting set be denoted by A_C^*. To avoid eavesdropping, Charlie adds l_2 randomly chosen decoys from $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ into A_C^* to get A_C^{**}. Charlie notes the positions and the initial states of these decoy photons and sends the resulting set A_C^{**} to Donald.
8) On receiving A_C^{**}, Donald interacts with Charlie in the similar manner as discussed in Step 6 to detect eavesdropping. If there is no such eavesdropping, Donald discards all the l_2 decoy photons to get A_C^*. In the following, Donald operates the quantum operators I or S or T to each photon of each group of A_C^* in the following manner: operates I to each of the photons of the ith group if $y_{i-1} = 0$, operates T to each of the photons of the ith group if $y_{i-1} = 1$ and $K(i) = 0$, and operates S to each of the photons of the ith group if $y_{i-1} = 1$, $K(i) = 1$. Suppose the resulting set is A_{CD}^*. In order to prevent eavesdropping, Donald adds l_3 decoy photons into A_{CD}^* to get A_{CD}^{**}. These decoy photons are randomly chosen from $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$. Donald notes the positions and initial states of these decoy photons and sends the resulting set A_{CD}^{**} to TP.
9) On receiving A_{CD}^{**}, TP interacts with Donald using similar technique as discussed in Step 6 to check whether there is eavesdropping. If not, then TP discards all l_3 decoy photons to get A_{CD}^*, and performs the following steps.
   a) Discards all the auxiliary photons added during Step 5 from each group and gets A_{CD} (say).
   b) Sets a count variable p as 0.
   c) Measures all the photons of the set A_{CD}.
   d) For $1 \leq i \leq r$, if the measurement result of the ith group of A_{CD} is same as the ith group of the set $A_{(3\pi/4)}$, then increases p by 1, i.e., $p = p + 1$.
   e) Writes L as the set of indices for which group the measurement result of A_{CD} matches with $A_{(3\pi/4)}$.
   f) If p is less than t, then aborts the process and outputs 0; otherwise, sends L to Charlie and Donald.
10) Let C_{int} and D_{int} be two empty sets. For each $i \in L$, Charlie computes $c_s = k^{-1}(i-1) \mod q$ and appends c_s to C_{int}, while Donald computes $d_t = k^{-1}(i-1) \mod q$ and appends d_t to D_{int}. Finally, the resulting sets C_{int} and D_{int} are, respectively, the desired intersection for Charlie and Donald. In fact $D_{\text{int}} = C_{\text{int}}$.

The communication flows of our proposed design are finally provided in Figs. 1–3. For better understanding of our design, a toy example is given in the Appendix to illustrate the operation of the proposed protocol. However, the proposed design is aimed at IoT-enabled privacy preserving ride sharing application, as described in Section VIII.

V. Correctness of QuTPSI

In the section, we prove the correctness of our proposed design. We first show that TP is able to correctly calculate the intersection cardinality. Without loss of generality, let the initial state of all the elements of the ith group of A prepared by TP in Step 4 be $|0'\rangle$. Recall that, TP also applies $\begin{pmatrix} 1 & 0 \\ 0 & e^{(i3\pi/4)} \end{pmatrix}$ on one of the copies of A to get $A_{(3\pi/4)}$. It implies that the state $|0'\rangle$ in $A_{(3\pi/4)}$ is rotated by an angle of $(3\pi/4)$ radian.

Consider the case when $K(i) = 0$. A similar analysis can be done for the other case $K(i) = 1$.

Case I ($K(i) = 0$)*:* The following possibilities may arise.

1) $x_{i-1} = 0$ *and* $y_{i-1} = 0$*:* Then Charlie applies quantum gate I on $|0'\rangle$, and subsequently Donald applies quantum gate I on $I|0'\rangle$. Therefore, in the resulting ith group, the states $|0'\rangle$s are rotated by an angle of zero radian. Thus, the measurement result does not match with the ith group of $A_{(3\pi/4)}$, and hence, it has no contribution to the intersection cardinality.
2) $x_{i-1} = 0$ *and* $y_{i-1} = 1$*:* In this case, Charlie and Donald operate I on $|0'\rangle$, and T on $I|0'\rangle$, respectively. Thereby, in the resulting ith group, the states $|0'\rangle$s are rotated by an angle of $(\pi/4)$ radians which mismatches with the ith group of $A_{(3\pi/4)}$. Therefore, it contributes nothing toward the intersection cardinality.
3) $x_{i-1} = 1$ *and* $y_{i-1} = 0$*:* Then, Charlie applies S on $|0'\rangle$, and later Donald operates quantum gate I on $S|0'\rangle$. As a consequence, in the resulting ith group, the states $|0'\rangle$s are rotated by an angle of $(\pi/2)$ radian. This result does not match with the ith group of $A_{(3\pi/4)}$. Thereby, it does not contribute anything toward the intersection cardinality.
4) $x_{i-1} = 1$ *and* $y_{i-1} = 1$*:* In this case, Charlie operates S on $|0'\rangle$, and subsequently Donald applies quantum gate T on $S|0'\rangle$. Therefore, in the resulting ith group, the states $|0'\rangle$s are rotated by an angle of $(3\pi/4)$ radians. So the resulting state will match with the ith group of $A_{(3\pi/4)}$. Note that, in this case TP increases p by 1 as mentioned in Step 9 of the protocol. Hence, it contributes toward intersection cardinality.

Case II ($K(i) = 1$)*:* A similar analysis can be done for this case with the following four possibilities.

1) $x_{i-1} = 0$ *and* $y_{i-1} = 0$*:* Then, the resulting ith group will not match with the ith group of $A_{(3\pi/4)}$ as the resulting ith group is obtained by an angle zero radian rotation.

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Consequently, this case contributes nothing toward the intersection cardinality.

2) $x_{i-1} = 0$ *and* $y_{i-1} = 1$*:* Here, the resulting ith group is obtained by $(\pi/2)$ radians rotation, where rotation of 0 radian is due to Charlie and $(\pi/2)$ radians rotation is due to Donald. As a result, it does not match with the ith group of $A_{(3\pi/4)}$. Thus, this case does not contribute to the intersection cardinality.
3) $x_{i-1} = 1$ *and* $y_{i-1} = 0$*:* In this case, due to Charlie's rotation about an angle of $(\pi/4)$ radians and Donald's rotation about an angle of 0 radian, the resulting ith group will not match with the ith group of $A_{(3\pi/4)}$ and hence, it will not provide any contribution to the intersection cardinality.
4) $x_{i-1} = 1$ *and* $y_{i-1} = 1$*:* In this case, Charlie and Donald gives $(\pi/4)$ radians and $(\pi/2)$ radians rotations, respectively. As a consequence, the resulting ith group is obtained by $(3\pi/4)$ radian rotation which is same as the ith group of $A_{(3\pi/4)}$. This case contributes to the intersection cardinality since TP increases p by 1 only when the ith resulting group matches with the ith group of $A_{(3\pi/4)}$.

From the above analysis, we observe that the initial state is rotated by an angle of $(3\pi/4)$ radian if and only if $x_{i-1} = 1$ and $y_{i-1} = 1$ for both cases $K(i) = 0$ and $K(i) = 1$. We now show that counting such pairs $(x_{i-1} = 1, y_{i-1} = 1)$ gives us the actual intersection cardinality, i.e., $|C \cap D| = |C^* \cap D^*| = \sum_{i=0}^{q-1} x_i y_i$. By the construction of (x_{i-1}, y_{i-1}), $i-1 \in C^* \cap D^*$ if $x_{i-1} = 1$ and $y_{i-1} = 1$. This yields $|C^* \cap D^*| = \sum_{i=0}^{q-1} x_i y_i$. Note that the elements of C^* and D^* are obtained from C and D, respectively, by modular multiplication with the integer k. For an $i \in C^* \cap D^*$ there exist $s \in \{1, \ldots, l\}$ and $t \in \{1, \ldots, m\}$ such that $(kc_s \mod q) = i$ and $(kd_t \mod q) = i$, i.e., $c_s = k^{-1}(i) \mod q$ and $d_t = k^{-1}(i) \mod q$ which implies $c_s = d_t \in C \cap D$. On the other hand, for all element $e \in C \cap D$, $ke \,(\mathrm{mod}\, q) \in C^* \cap D^*$. Thus, we can conclude that $|C \cap D| = |C^* \cap D^*|$ which is equal to $\sum_{i=0}^{q-1} x_i y_i$.

We now show that Charlie and Donald are able to calculate the intersection correctly. If $i \in L$ then $|0'\rangle$ is rotated by $(3\pi/4)$ radians. It means that $x_{i-1} = 1$ and $y_{i-1} = 1$, which in turn implies that $i-1 \in C^* \cap D^*$. Hence, there exist $s \in \{1, \ldots, l\}$ and $t \in \{1, \ldots, m\}$ such that $(kc_s \mod q) = i-1$ and $(kd_t \mod q) = i-1$, i.e., $c_s = k^{-1}(i-1) \mod q$ and $d_t = k^{-1}(i-1) \mod q$. Thus, $c_s = d_t \in C \cap D$. In other words, corresponding to each $i \in L$ or $i-1 \in C^* \cap D^*$, Charlie and Donald are able to compute the corresponding $e \in C \cap D$ such that $e = k^{-1}(i) \mod q$. Again, $|C \cap D| = |C^* \cap D^*|$. Hence, the participants can correctly calculate the intersection $C \cap D$ by computing $k^{-1}(i-1) \mod q$ for each $i \in L$.

VI. Security Analysis

In this section, we prove that QuTPSI protocol meets the following security requirements.

1) TP cannot obtain any information about the private sets of Charlie and Donald.
2) Donald gets no information about the private set of Charlie apart from $C \cap D$.
3) Charlie does not get any information about Donald's private set, beyond $C \cap D$.
4) An outsider is unable to obtain information about the private sets of Charlie and Donald.

Theorem 1: TP cannot obtain any information about the private sets of Charlie and Donald.

Proof: A dishonest TP may prepare entangled photons instead of single photons to obtain information about the private sets of Charlie and Donald. Suppose TP initially prepares the entangled photon $(1/\sqrt{3})|01\rangle + (\sqrt{2}/\sqrt{3})|10\rangle$, keeps the first photon and sends the second photon to Charlie. Then, the reduced density matrix corresponding to the first photon is

$$\frac{1}{3}|0\rangle\langle 0| + \frac{2}{3}|1\rangle\langle 1|.$$

As the reduced density matrix is independent of the unitary operator applied to the photon, so TP cannot gain any information about the private set of Charlie.

There is another possibility of how TP can act. If TP eavesdrops to get information about the private sets of Charlie, then he may apply the unambiguous state discrimination (USD) to successfully guess the initial states of the selected photons which is different from the decoy photons. The success probability to know the initial states is $\mathsf{Prob}^{\mathrm{USD}} = 1 - F(\rho_0, \rho_1)$, where $F(\rho_0, \rho_1)$ is the fidelity between the two quantum states which TP seeks to discriminate. Let $|0'\rangle$ be the initial state prepared by TP. Then, $S(|0'\rangle) = \cos(\theta)|0\rangle + e^{(i\pi/2)}\sin(\theta)|1\rangle$ (i.e., $x_{i-1} = 1$), and thus, the successful probability of USD is

$$\begin{aligned}\mathsf{Prob}^{\mathrm{USD}} &= 1 - F(\rho_0, \rho_1)\\ &= 1 - \left(\cos^2(2\theta) + \frac{1}{4}\sin^4(2\theta)\right).\end{aligned}$$

If $\theta = (\pi/20)$

$$\begin{aligned}\mathsf{Prob}^{\mathrm{USD}} &= 1 - \left(\cos^2(2\theta) + \frac{1}{4}\sin^4(2\theta)\right)\\ &= 1 - \left(\cos^2\left(\frac{2\pi}{20}\right) + \frac{1}{4}\sin^4\left(\frac{2\pi}{20}\right)\right)\\ &= 1 - 0.906\\ &\approx 0.094.\end{aligned}$$

Therefore, the success probability of TP in deducing $x_{i-1} = 0$ or $x_{i-1} = 1$ is 0.094. Also, TP does not know the secret key k, and hence, TP cannot obtain any information about the private set of Charlie. ■

Theorem 2: Donald gets no information about the private set of Charlie.

Proof: In order to obtain any information about the private set of Charlie, Donald measures the photons of the set A_C^*. If $x_{i-1} = 0$ then Charlie did nothing and therefore the ith group is the same as what TP prepared initially. As Donald does not know the actual measurement bases, he can guess $x_{i-1} = 0$ with probability $(1/2)$. If $x_{i-1} = 1$, then Charlie applies S or T according to $K(i)$. Also, in this case, Donald guesses $x_{i-1} = 1$ with probability $(1/2)$ as he does not know the actual measurement bases. As a result, Donald is unable to extract any information about the private set of Charlie. ■

Theorem 3: Charlie does not get any information about Donald's private set, beyond $C \cap D$.

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TABLE V
COMPARISON TABLE OF OUR PROPOSED PROTOCOL WITH EXISTING TPSI PROTOCOLS

Protocol	Computation cost	Communication cost	Quantum cryptography based	Long-term security
Hallgren *et al.* [47]	$\mathcal{O}(n^2)$	$\mathcal{O}(\lambda n)$	No	No
Ghosh and Nilges [48]	$\mathcal{O}(n \log^2 n)$	$\mathcal{O}(n\lambda)$	No	No
Ghosh and Simkin [8]	$\mathcal{O}((n-t)^4)$	$\mathcal{O}(t)$	No	No
Badrinarayanan *et al.* [10]	$\mathcal{O}((n-t)^4)$	$\mathcal{O}(Nt)$	No	No
Zhang *et al.* [44]	$\mathcal{O}(n \log n)$	$\mathcal{O}(\lambda M)$	No	No
Proposed (QuTPSI)	$\mathcal{O}(q+(r+r^*)q)$	$\mathcal{O}(q+(r+r^*)q)$	Yes	Yes

n: size of the private sets, λ: security parameter, δ: the number of hash functions, t: the threshold value, M: the length of Bloom filter and N: the number of parties, q: size of the private vectors of the users, r: number of initial photons, r^*: number of auxiliary photons.

Proof: If Charlie wants to get any information about the private set of Donald, he needs to intercept the quantum communication channel between Donald and TP. Note that, Charlie does not know the actual positions of the initial photons and auxiliary photons. In addition, all these photons are nonorthogonal, therefore, these are indistinguishable. Hence, Charlie does not get any information about Donald's private set, beyond $C \cap D$. ■

Theorem 4: An outsider is unable to obtain any information about the private sets of Charlie and Donald.

Proof: If an outsider wants any information about the private sets of Charlie and Donald then he may perform entangle measurement attack on a decoy photon state, say $|\psi\rangle_d \in \{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$. On receiving the decoy state $|\psi\rangle_d$, he prepares an ancillary state $|0\rangle_a$ and operates an oracle operator $\mathcal{O}_\delta : |\delta_1\rangle|\delta_2\rangle \mapsto |\delta_1\rangle|\delta_2 \oplus \delta(\delta_1)\rangle$.

1) *Case I:* If $|\psi\rangle_d$ is $|0\rangle$ or $|1\rangle$, then

$$\mathcal{O}_\delta|\psi\rangle_d|0\rangle_a = \begin{cases} |0\rangle|0 \oplus \delta(0)\rangle_a \\ = |0\rangle|\delta(0)\rangle_a, \quad \text{if} |\psi\rangle_d = |0\rangle \\ |1\rangle|0 \oplus \delta(1)\rangle_a \\ = |1\rangle|\delta(1)\rangle_a, \quad \text{if} |\psi\rangle_d = |1\rangle. \end{cases}$$

2) *Case II:* If $|\psi\rangle_d \in \{|+\rangle, |-\rangle\}$, then

$$\begin{aligned} \mathcal{O}_\delta|\psi\rangle_d|0\rangle_a &= \frac{\mathcal{O}_\delta|0\rangle|0\rangle_a \pm \mathcal{O}_\delta|1\rangle|0\rangle_a}{\sqrt{2}} \\ &= \frac{|0\rangle|0 \oplus \delta(0)\rangle_a \pm |1\rangle|0 \oplus \delta(1)\rangle_a}{\sqrt{2}} \\ &= \frac{|0\rangle|\delta(0)\rangle_a \pm |1\rangle|\delta(1)\rangle_a}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}}\left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|\delta(0)\rangle_a \pm |\delta(1)\rangle_a}{\sqrt{2}} \right. \\ &\quad \left. + \frac{|0\rangle - |1\rangle}{\sqrt{2}} \otimes \frac{|\delta(0)\rangle_a \mp |\delta(1)\rangle_a}{\sqrt{2}}\right]. \end{aligned}$$

By the above analysis, we see that if $|\psi\rangle_d \in \{|0\rangle, |1\rangle\}$, the outsider can guess correctly but if $|\psi\rangle_d \in \{|+\rangle, |-\rangle\}$ then the success probability is $(1/2)$. In addition, all these photons are non orthogonal so these states are indistinguishable. Therefore, the outsider fails to obtain any information about the private sets of Charlie or Donald. Also, an outsider does not know the secret key k, and hence, the outsider cannot obtain any information about the entities private sets. ■

VII. PERFORMANCE ANALYSIS

The communication cost and computation cost of the proposed scheme are discussed below. Next, we compare the performance of the proposed scheme with the existing competing schemes of Hallgren et al. [47], Ghosh, and Nilges [48], Ghosh and Simkin [8], Badrinarayanan et al. [10], and Zhang et al. [44].

A. Communication Cost

In the proposed scheme, $\lceil \log_2(q) \rceil + q$ qubits are required to be transferred during the key sharing. $3(r + r^*)q + l_1 + l_2 + l_3$ qubits are needed to be shared for the computation of cardinality of private sets. Let $|L|$ be the cardinality of L in bits. The set of indices L can be sent as a bitstring of length q by putting 1 at the places for which the indices are contained in L and 0 at the remaining places. Thus, $2q$ additional bits are needed to be transferred for the computation of intersection.

B. Computation Cost

In the proposed scheme, $\lceil \log_2(q) \rceil + q + l_1 + l_2 + l_3$ projective measurements are required. $2rq + r^*q + l_1 + l_2 + l_3$ single photons are needed to be prepared. $2rq + 2(r + r^*)q$ quantum unitary operations have to be performed. In addition, we also require $l + m + 2p$ modular multiplications.

Our scheme is very efficient as it does not employ any complicated oracle operators. In addition, it utilizes single photons and simple single-particle projective measurements. Thereby, it has the potential of being implemented with the current quantum hardware technologies [49], [50].

C. Comparative Study

Since our proposed design is the first QC-based TPSI, we compare our scheme with other existing state-of-the-art TPSI protocols. A comparative study on communication and computational costs, and functionality features among the proposed scheme and the existing competing schemes of Hallgren et al. [47], Ghosh and Nilges [48], Ghosh and Simkin [8], Badrinarayanan et al. [10], and Zhang et al. [44], has been provided in Table V.

Our proposed design is in quantum domain, and hence, its computational and communication overheads involve qubits and other quantum resources. The existing competing schemes [8], [10], [44], [47], [48] have classical computational and communication costs. Therefore, no direct comparison can be made based on computational and communication costs point of view. However, our proposed design is superior to these schemes due to its advanced security features. The design of the schemes [8], [10], [44], [47], [48] are based on the

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classical hardness assumptions. Hence, in near future, these scheme may become obsolete and insecure. Even a single quantum computer is enough to mount an attack over these schemes. These schemes also fail to provide long-term security. In contrast, our proposed design is long-term secure. In addition, it provides security against quantum adversary as well as classical adversary. Since security is one of the most important component of an IoT system, our proposed design assures strong security guarantees with minimal quantum resources. The results are summarized in Table V.

VIII. Application to IoT-Enabled Privacy-Preserving Ride Sharing

In this section, we explain how our proposed quantum secure TPSI scheme (QuTPSI) can be utilized as a building block in an IoT-enabled ride sharing setting.

We first give a high-level overview of the system design. Our model allows two people, say Charlie and Donald, to share the ride if the intersection of their routes is greater than or equal to a certain predefined threshold value. The complete system model can be categorized into four phases. In Phase I, Charlie and Donald interact with each other. Here, they execute a QKD among themselves to hide their private trajectories. The next two phases (i.e., Phases II and III) involve a trusted TP, which assists Charlie and Donald in calculating the intersection of their private routes. At the end of Phase III, Charlie and Donald know what their shared path is. In Phase IV, the clients who wish to share the ride, contact a car leasing service provider Cab for the further process.

A. Various Phases

The details of the phases are provided in the following sections. A supporting example illustrating the operation of Phases I, II, and III are also provided in the Appendix.

1) Phase I: Let Charlie and Donald be the two clients who wish to share their journey. Suppose $\mathsf{traj_C}$ and $\mathsf{traj_D}$ denote, respectively, the private trajectories of Charlie and Donald. Now, they execute a QKD protocol to share a nonzero binary secret key, say sk. In the following, they use sk to modify $\mathsf{traj_C}$ and $\mathsf{traj_D}$ into $\mathsf{traj_C^*}$ and $\mathsf{traj_D^*}$, respectively. Using the aforementioned modifications, Charlie and Donald then generate respective private vectors corresponding to their trajectories, namely, $\mathbf{x}$ and $\mathbf{y}$. In the end, Charlie and Donald share a q-bit private-key K by again employing some QKD protocol. They store this key K for the future purposes. An illustrative diagram for this purpose is already given in Fig. 1.

2) Phase II: This phase involves the trusted TP. TP randomly chooses a set A consisting of q groups of photons, where each group contains r photons. It makes two such copies of A, then applies $R_{3\pi/4}$ on one such copy, and writes down the resulting set as $A_{3\pi/4}$. It keeps $A_{3\pi/4}$ and proceeds with A in the next step. In each group of A, TP randomly inserts randomly chosen single photons, also known as auxiliary photons. The resulting set is denoted by A^*. In addition, TP also randomly inserts randomly selected decoy photons in A^* to get A^{**}. After noting down the position and initial states of each photon, TP sends A^{**} to Charlie through a quantum channel.

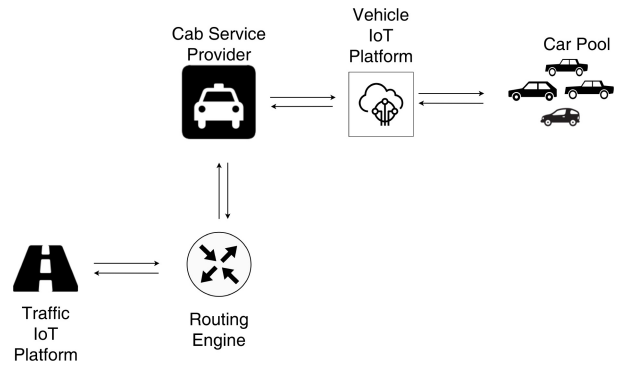

Fig. 4. Communication flow of Phase IV.

Charlie on his end, discards the decoy photons from A^{**}. Now, Charlie uses the key K shared in Phase I and operates the quantum operators I or S or T to each photon of all the groups of A^* as per the rules specified in Step 7 of the proposed scheme (see Section IV-B). To avoid eavesdropping, Charlie also adds some randomly chosen decoy photons to form a set A_C^{**}, and sends the final set to Donald. On receiving A_C^{**} from Charlies, Donald proceeds in the similar fashion as Charlie. In particular, he discards all the decoy photons, and then uses the preshared key K to operate the quantum operators I or S or T to each photon of each group according to rules specified in Step 8. In order to prevent eavesdropping, Donald adds decoy photons to form the set A_{CD}^{**}. Donald sends the resulting set A_{CD}^{**} to TP. In the end, TP follows Step 9 to calculate L, and sends it to Charlie and Donald. L has sufficient information which can be used by Charlie and Donald to calculate the intersection. Fig. 2 gives an illustrative summary of this phase.

3) Phase III: For each $i \in L$, Charlie computes $c_s = k^{-1}(i-1) \mod q$, while Donald computes $d_t = k^{-1}(i-1) \mod q$. This collection of c_s and d_s actually gives them the shared trajectories. In other words, $\{c_s\} = \mathsf{traj_C} \cap \mathsf{traj_D} = \{d_t\}$ (refer to Section V).

4) Phase IV: Charlie and Donald know what paths they are going to share. They contact a car-leasing service provider (we call it as Cab) for the ride-sharing service. A message broker is used to send their requests to the Cab. Cars associated with the Cab exchange data with it through a customized vehicle IoT platform. These cars have IoT sensors fitted in that collect data related to location of the car, pick-ups, and drop-offs, and send it to the vehicle IoT platform. While on the other side, Cab sends the information about client-assignments and route-instructions to the connected cars via the same vehicle IoT platform. Through the data sent by a traffic routing engine, Cab calculates the travel times for the shared trajectories. The data received through routing engine can also be utilized to provide a better service to the users. This routing engine gets information about changes in traffic situation through a dedicated traffic-IoT platform. Fig. 4 shows an illustrative diagram for this phase.

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B. Discussions

We have seen how our scheme (QuTPSI) can be utilized in ride-sharing setting. We now discuss some of the security risks and how our scheme alleviates the security concerns. Charlie and Donald might try to know the private trajectories of each other and hamper the security and privacy of the system. Our design ensures that the passengers learn not more than the intersection of their private trajectories (refer to Theorems 2 and 3). The TP involved in Phases II and III cannot obtain any information about the private routes of passengers (refer to Theorem 1). Our design also ensures that no outsider can learn the passengers private information (refer to Theorem 4). Our scheme relies on QC. As a consequence, it provides security against the attacks by quantum computer and achieves long-term security. Thus, QuTPSI can be utilized as a building block to design a quantum secure IoT-enabled ride sharing scheme.

IX. Conclusion

In this article, to the best of our knowledge, we propose the *first* QC-based TPSI protocol QuTPSI. The security of QuTPSI is based on the quantum physical laws. Hence, it provides security against attacks through quantum algorithms. In addition, it also provides long-term security, unlike the classical TPSI protocols. Moreover, we investigated the possible application of our presented design QuTPSI in the domain of ride sharing to develop a quantum computer resistant long-term secure privacy-preserving ride sharing application.

Appendix
Toy Example

In this section, we give an illustrative example for our proposed design.

A. Phase I

Let $\mathsf{traj_C}$ and $\mathsf{traj_D}$ be the private trajectories of Charlie and Donald, represented by the set $C = \{1, 2, 4\}$ and $D = \{0, 1, 2, 3\}$, respectively. The individual elements of these sets belong to $\mathbb{Z}_5$. Charlie and Donald share a secret integer $k = 2$ through the QKD protocol. We set the threshold value to be $t = 2$. Now, using $k = 2$, they transform the sets corresponding to their private trajectories into $\mathsf{traj}^*_\mathsf{C} = \{2 \cdot 1 \mod 5, 2 \cdot 2 \mod 5, 2 \cdot 4 \mod 5\} = \{2, 4, 3\}$ and $\mathsf{traj}^*_\mathsf{D} = \{2 \cdot 0 \mod 5, 2 \cdot 1 \mod 5, 2 \cdot 2 \mod 5, 2 \cdot 3 \mod 5\} = \{0, 2, 4, 1\}$, respectively. Utilizing the sets $\mathsf{traj}^*_\mathsf{C}$ and $\mathsf{traj}^*_\mathsf{D}$, Charlie and Donald generate their respective private vectors $x = 00111$ and $y = 11101$ by going through the rules described in Step 2. In the next step, by employing the QKD protocol, they share a 5-bit secret key $K = K_1K_2K_3K_4K_5 = 01011$.

B. Phase II

TP prepares five groups of photons with $r = 3$, and $r^* = 2$. *TP* then prepares two copies of the set $A = \{|0'\rangle, |0'\rangle, |0'\rangle; |+'\rangle, |+'\rangle, |+'\rangle; |0'\rangle, |0'\rangle, |0'\rangle; |1'\rangle, |1'\rangle, |1'\rangle; |-'\rangle, |-'\rangle, |-'\rangle\}$. In the following, *TP* applies $\begin{pmatrix} 1 & 0 \\ 0 & e^{(i3\pi/4)} \end{pmatrix}$ on one of the copies and gets the set $A_{(3\pi/4)}$. In each group of A, *TP* randomly inserts $r^* \leq r$ that are randomly chosen single photons (also called auxiliary photons). Let the position of the auxiliary photons in each group be the following: $\text{Group}_1 =$ 4th and 5th positions, $\text{Group}_2 =$ 4th and 5th positions, $\text{Group}_3 =$ 1st and 2nd positions, $\text{Group}_4 =$ 1st and 5th positions, and $\text{Group}_5 =$ 3rd and 5th positions. Let the resulting set be $A^* = \{|0'\rangle, |0'\rangle, |0'\rangle, |1'\rangle, |0'\rangle; |+'\rangle, |+'\rangle, |+'\rangle, |1'\rangle, |-'\rangle; |1'\rangle, |0'\rangle, |0'\rangle, |0'\rangle, |0'\rangle; |1'\rangle, |1'\rangle, |1'\rangle, |1'\rangle, |1'\rangle; |-'\rangle, |-'\rangle, |1'\rangle, |-'\rangle, |+'\rangle\}$. In the next step, *TP* adds some decoy photons in A^* to get A^{**} and sends A^{**} to Charlie. Charlie discards decoy photons from A^{**} to get A^*. Now, Charlie operates the quantum operators I or S or T on each photon of all the group of A^* according to rules described in Step 7. For our case, Table VI summarizes which operator to use.

Let the resulting set be denoted by $A^*_C = \{|0'\rangle, |0'\rangle, |0'\rangle, |1'\rangle, |0'\rangle; |+'\rangle, |+'\rangle, |+'\rangle, |1'\rangle, |-'\rangle; S|1'\rangle, S|0'\rangle, S|0'\rangle, S|0'\rangle, S|0'\rangle; T|1'\rangle, T|1'\rangle, T|1'\rangle, T|1'\rangle, T|1'\rangle; T|-'\rangle, T|-'\rangle, T|1'\rangle, T|-'\rangle, T|+'\rangle\}$. Charlie further adds decoy photons to A^*_C to get A^{**}_C and sends it to Donald. Donald removes the decoy photons from A^{**}_C to get A^*_C. In the following, Donald operates the quantum operators I or S or T on each photon of each group of A^*_C by following the rules described in Step 8. Table VII describes the choice of operator for our case.

Let the resulting set be $A^*_{CD} = \{T|0'\rangle, T|0'\rangle, T|0'\rangle, T|1'\rangle, T|0'\rangle; S|+'\rangle, S|+'\rangle, S|+'\rangle, S|1'\rangle, S|-'\rangle; TS|1'\rangle, TS|0'\rangle, TS|0'\rangle, TS|0'\rangle, TS|0'\rangle; T|1'\rangle, T|1'\rangle, T|1'\rangle, T|1'\rangle, T|1'\rangle; ST|-'\rangle, ST|-'\rangle, ST|1'\rangle, ST|-'\rangle, ST|+'\rangle\}$ In the following, Donald adds the decoy photons to A^*_{CD} to get A^{**}_{CD} and sends the resulting set to *TP*. *TP* removes the decoy photons from A^{**}_{CD} to get A^*_{CD}. It discards auxiliary photons (underlined states) from $A^*_{CD} = \{T|0'\rangle, T|0'\rangle, T|0'\rangle, \underline{T|1'\rangle}, \underline{T|0'\rangle}; S|+'\rangle, S|+'\rangle, S|+'\rangle, \underline{S|1'\rangle}, \underline{S|-'\rangle}; \underline{TS|1'\rangle}, \underline{TS|0'\rangle}, \underline{TS|0'\rangle}, TS|0'\rangle, TS|0'\rangle; \underline{T|1'\rangle}, \underline{T|1'\rangle}, \underline{T|1'\rangle}, \underline{T|1'\rangle}, \underline{T|1'\rangle}; ST|-'\rangle, ST|-'\rangle, ST|1'\rangle, \underline{ST|-'\rangle}, ST|+'\rangle\}$ to get $A_{CD} = \{T|0'\rangle, T|0'\rangle, T|0'\rangle; S|+'\rangle, S|+'\rangle, S|+'\rangle; TS|0'\rangle, TS|0'\rangle, TS|0'\rangle; T|1'\rangle, T|1'\rangle, T|1'\rangle; ST|-'\rangle, ST|-'\rangle, ST|-'\rangle\}$. In the following, *TP* measures all photons of all groups and compare them with $A_{(3\pi/4)}$. Since $TS = ST = \begin{pmatrix} 1 & 0 \\ 0 & e^{(i3\pi/4)} \end{pmatrix}$, therefore 3rd and 5th groups of A_{CD} are identical with the 3rd and 5th groups of $A_{(3\pi/4)}$. As two positions are matching, the count variable p is equal to 2. Note that, if the ith group matches,

TABLE VI
Choice of Operator for Charlie in Toy Example

x	0	0	1	1	1
K	0	1	0	1	1
Operator	I	I	S	T	T

TABLE VII
Choice of Operator for Donald in Toy Example

y	1	1	1	0	1
K	0	1	0	1	1
Operator	T	S	T	I	S

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then $i-1 \in L$. Then, $L = \{2, 4\}$. As $p \geq t$, *TP* sends L to both Charlie and Donald.

C. Phase III

Charlie computes $k^{-1} = 2^{-1} \pmod 5 = 3$ and as $L = \{2, 4\}$ the desired intersection $C \cap D = \{2.3 \pmod 5, 4.3 \pmod 5)\} = \{1, 2\}$. In a similar way, Donald computes the intersection as $C \cap D = \{1, 2\}$.

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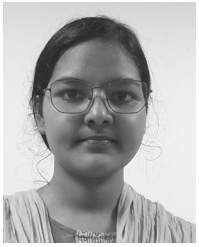

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